

Algebra Basics

AI Tutor - Mathematics - Lesson 2

Introduction

Algebra is the branch of mathematics that uses symbols, usually letters like x and y , to represent numbers we don't yet know. An equation is a statement that two expressions are equal.

Explanation

Algebra is the branch of mathematics that uses symbols, usually letters like x and y , to represent numbers we don't yet know. An equation is a statement that two expressions are equal, and 'solving' an equation means finding the value of the unknown variable that makes the statement true. The key technique is keeping both sides of the equation balanced: whatever operation you perform on one side, you must perform on the other.

Key Points

- A variable is a symbol representing an unknown value.
- An equation states that two expressions are equal.
- Solving an equation means isolating the variable using balanced operations.
- PEMDAS defines the order of operations: Parentheses, Exponents, Multiplication/Division, Addition/Subtraction.

Examples

- $x + 5 = 12$ -> subtract 5 from both sides -> $x = 7$
- $3x = 21$ -> divide both sides by 3 -> $x = 7$
- $x / 2 = 4$ -> multiply both sides by 2 -> $x = 8$

Summary

Algebra represents unknown numbers with variables and solves equations by performing the same operation on both sides to isolate the variable.

Practice Questions

- 1 Solve for x : $x + 8 = 15$
- 2 Solve for x : $4x = 32$
- 3 Solve for x : $2x + 3 = 11$

Quiz

- Q1.** What is the value of x in ' $x + 5 = 12$ '?
a) 5 | b) 7 | c) 12 | d) 17

Q2. What operation isolates x in ' $3x = 21$ '?

a) Add 3 | b) Subtract 21 | c) Divide by 3 | d) Multiply by 3

Q3. In PEMDAS, what comes right after Parentheses?

a) Addition | b) Multiplication | c) Exponents | d) Subtraction